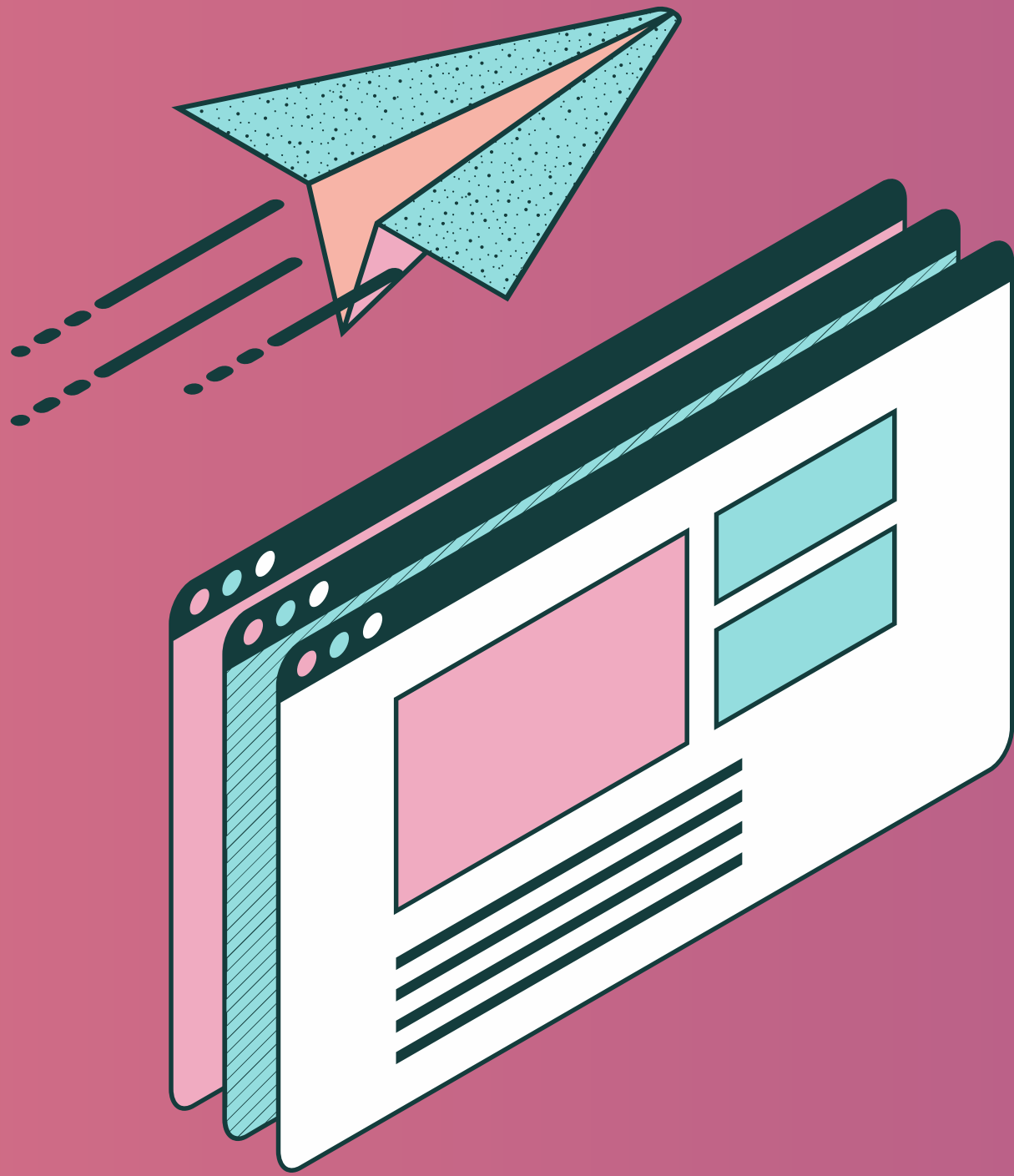


Image detection and repairing

112502024 張凱翔

112502564 謝坤霖



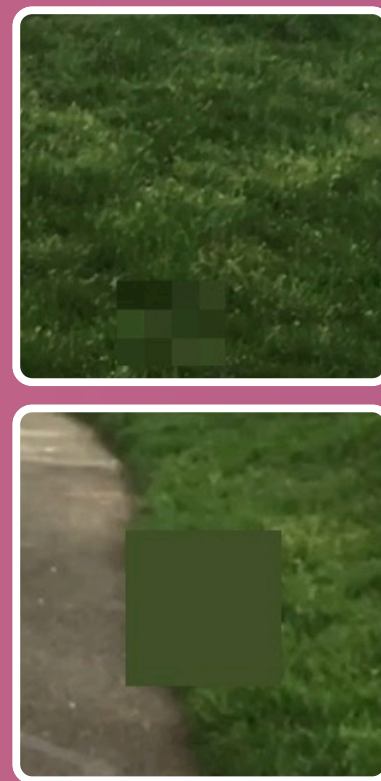


Goal

- Perform multi-class classification based on the condition of damaged images.
- Detect corrupted locations by incorporating bounding boxes.
- Perform various image restoration according to different corruption type

Multi-class classification

- Blocks
- Ghosting
- Mosaic
- Tear
- Normal

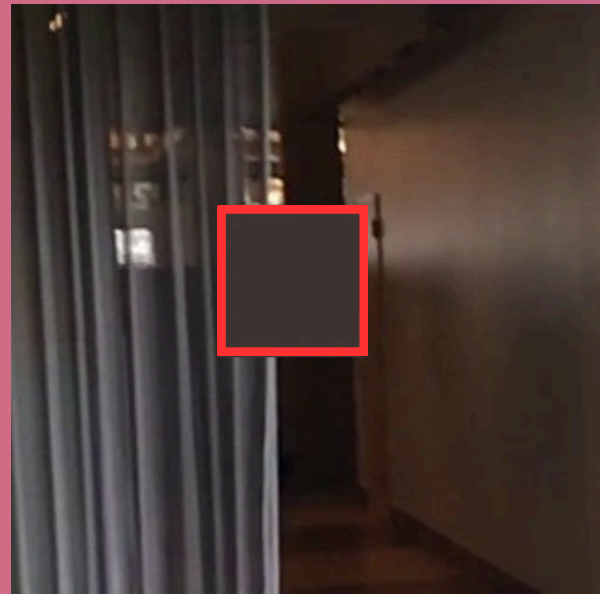


Bounding box coordinates

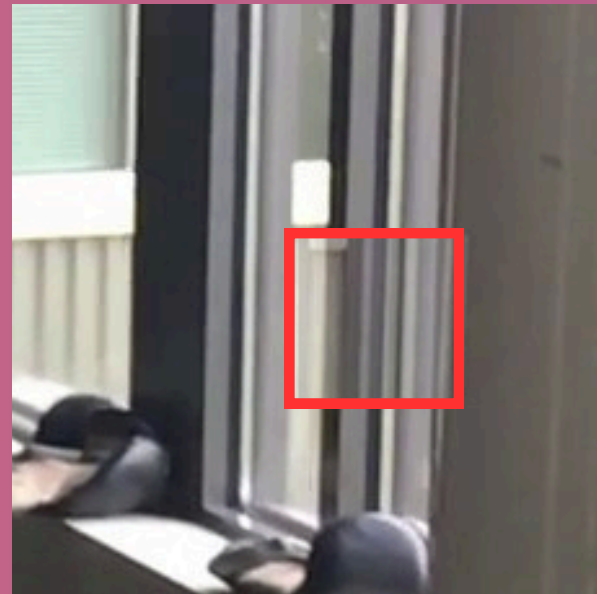
(x, y, w, h)

- X-coordinate
- Y-coordinate
- weidth
- height

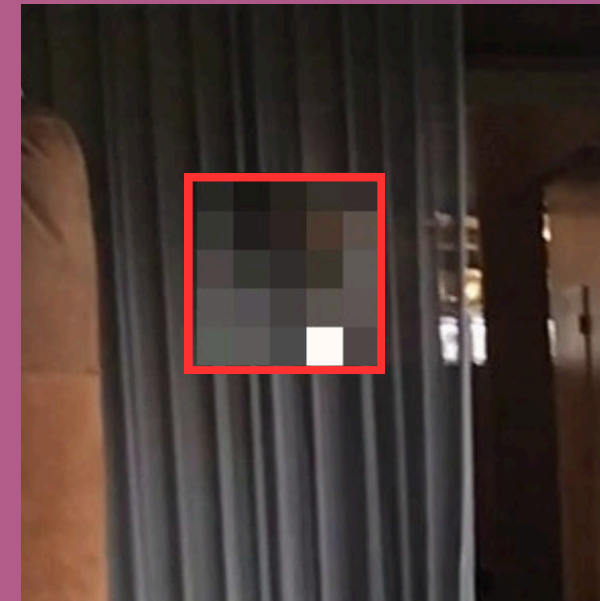
Corruption Types



Blocks



Ghosting



Mosaic



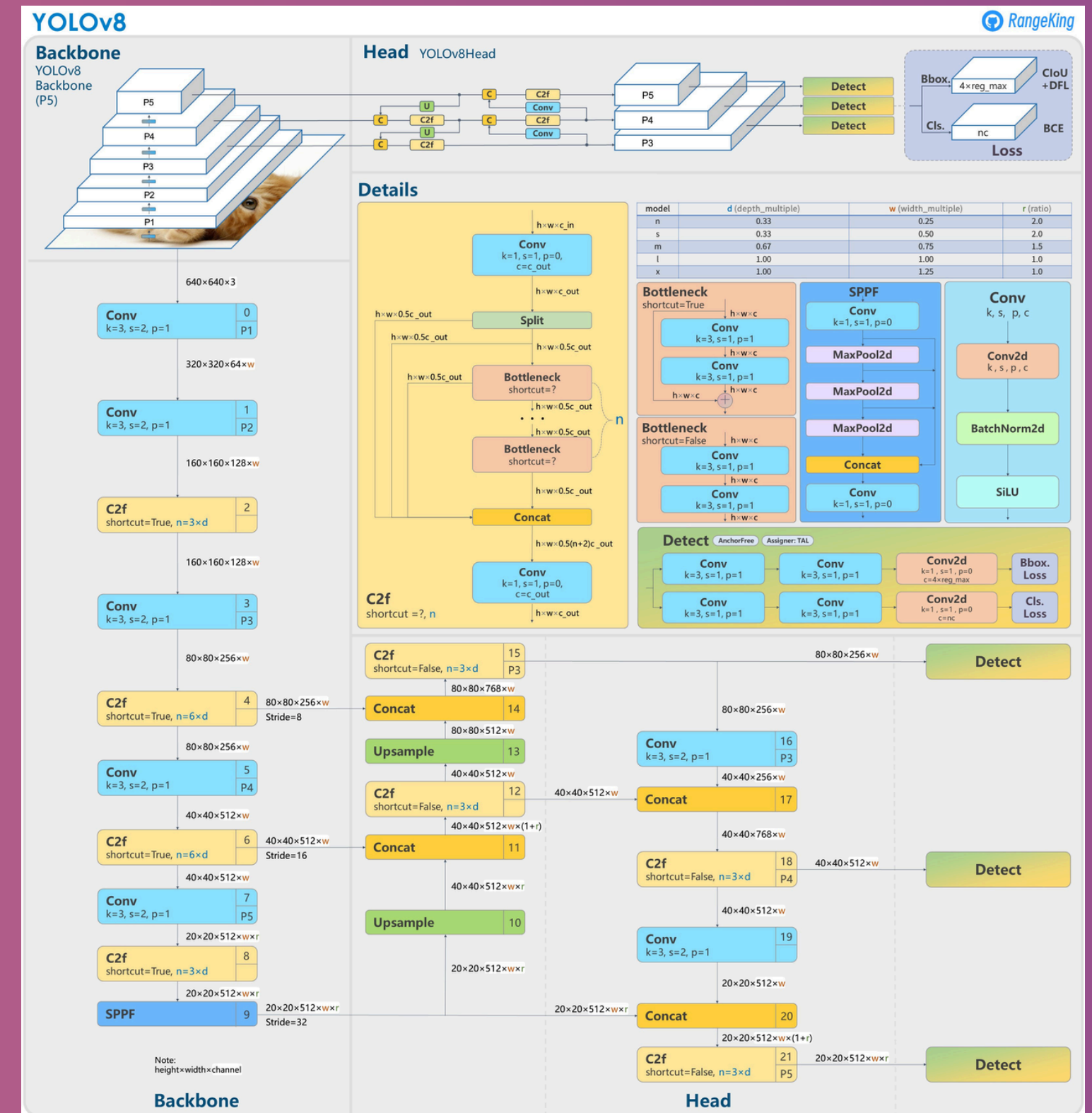
Tear

Implementation

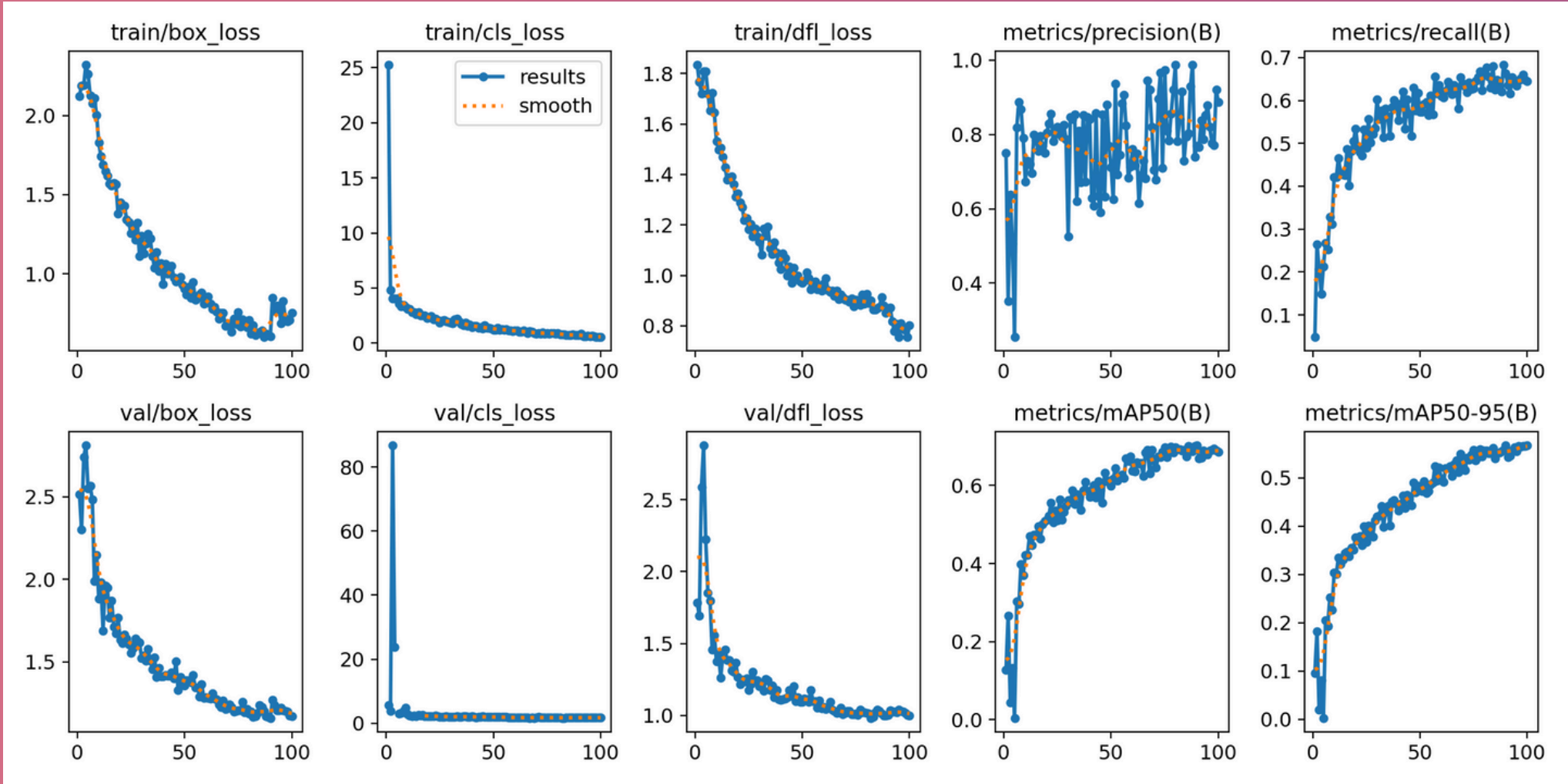
Split the dataset : train 80%, val 10%, test 10%

Model choose

- Architecture Components:
 - Backbone: Feature extraction
 - Head: Multi-scale detection
- Model Variant: YOLOv8s
- Scaling Parameters:
 - Depth scaling factor (d) = 0.33
 - Width scaling factor (w) = 0.50
- Advantage: Balance between model size and detection performance



Training Result & Comparison



Train	epochs
cls_loss	decrease from 25 to nearly 0 in 100 epochs
dfl_loss	decrease from 1.8 to nearly 0 in 100 epochs

Validation	epochs
cls_loss	decrease sharply from 25 to nearly 0 in few epochs
dfl_loss	decrease from 3.0 to nearly 0 in 100 epochs

Detection Output

Blocks



Mosaic



Ghosting



Tear



Image Repairing

Operation Principle of Deep Image Prior (DIP)

1. No Training Data Required

Unlike traditional deep learning methods that require massive datasets, DIP operates on a single corrupted image.

2. Network Structure as the Prior

The core idea is that the structure of a CNN itself acts as a "prior." CNNs are naturally good at generating natural image statistics but are resistant to learning random noise.

3. The Optimization Process

The process starts with a randomly initialized network and a fixed random noise input. The network is trained to reproduce the corrupted target image.

4. Early Stopping

Since the network learns meaningful structures (the clean image) much faster than it memorizes noise, we stop the training process before it overfits to the corruption. The result at this intermediate step is the restored, clean image.



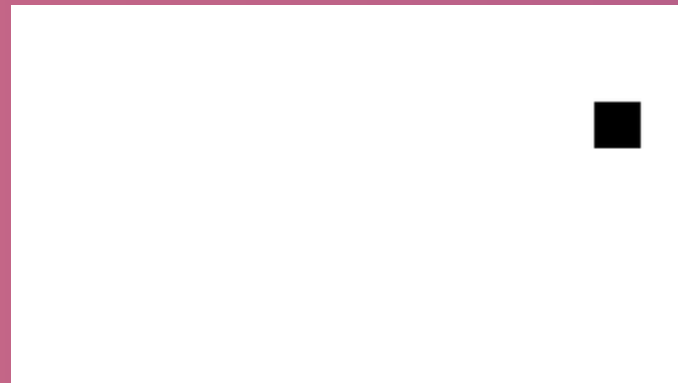
DIP method

1. Inpainting

The key mechanism is the application of a **binary mask** within the loss function, which restricts the optimization to only the known, undamaged pixels. Due to the inductive bias of the CNN structure, the network automatically propagates valid textures from the surroundings into the masked gaps.



original graph



mask

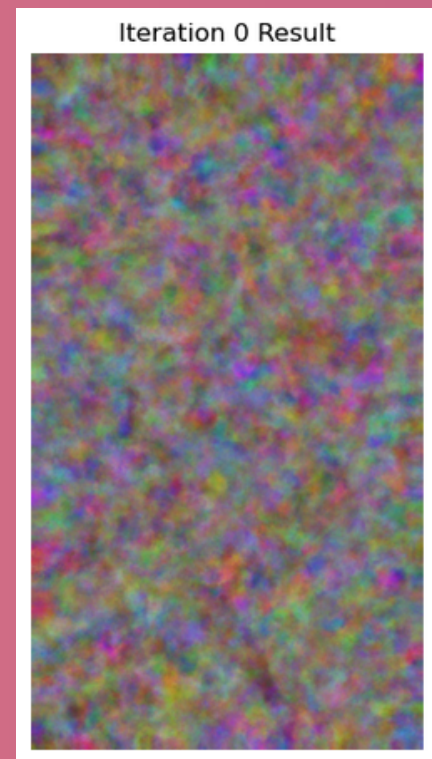
2. Super resolution

The optimization process involves a differentiable **downsampling** operator.

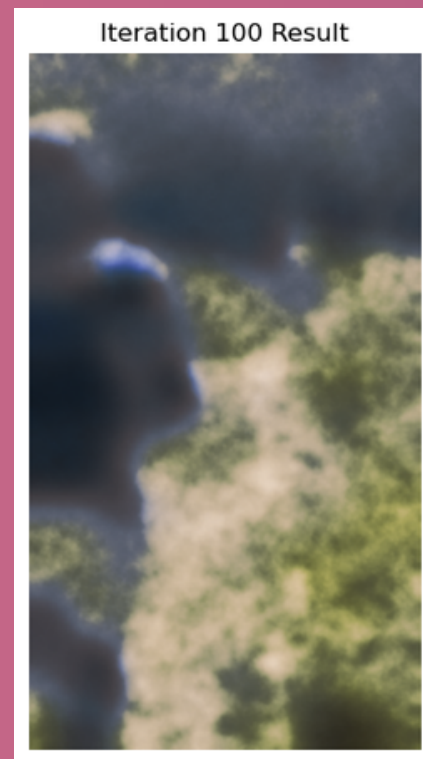
In each iteration, the network's high-resolution output is downsampled to match the dimensions of the original low-resolution input.

Among possible high-resolution images, the CNN structure converges to the one with the most reasonable natural details and sharp edges.

Generating process



Initial state
(Iteration 0)



Iteration 100



Iteration 300



The best output
Stop here!



Final state
(overfitting)



Restored Result

Inpainting(Block)

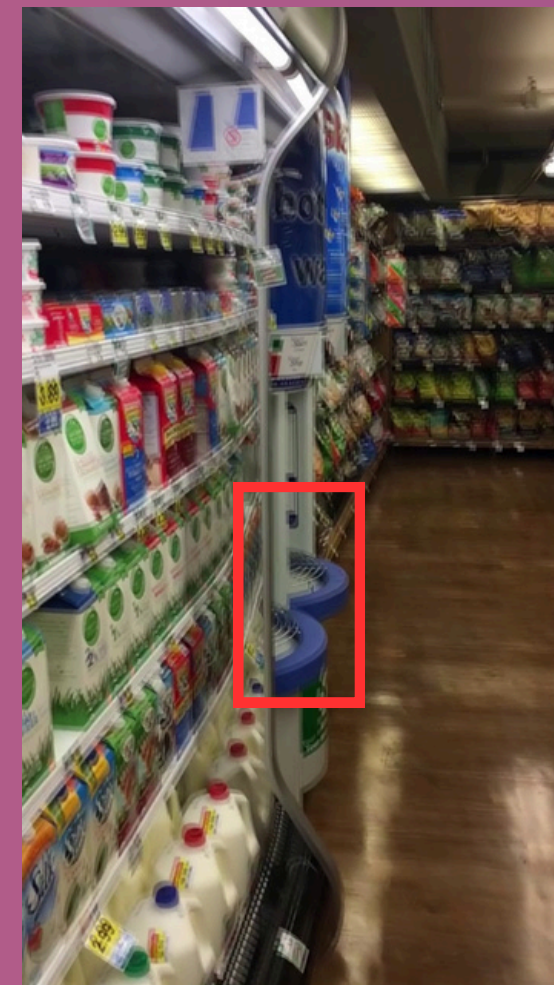


Original graph

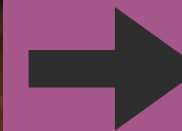


restored graph

Super resolution(Ghosting)



Original graph



restored graph

Tear solution

Image Registration

- Phase Correlation
- Canny Edge Detection
- Affine Transformation



Image Inpainting

- Pixel Copying
- Inpainting



Image Registration & Inpainting

1. Phase Correlation

- Detects horizontal offset between image segments
- Uses frequency domain correlation to calculate displacement

2. Canny Edge Detection

- Used when phase correlation response is low
- Assists in improving offset detection accuracy

3. Affine Transformation

- Translates offset segments back to correct positions
- Uses translation matrix for geometric correction

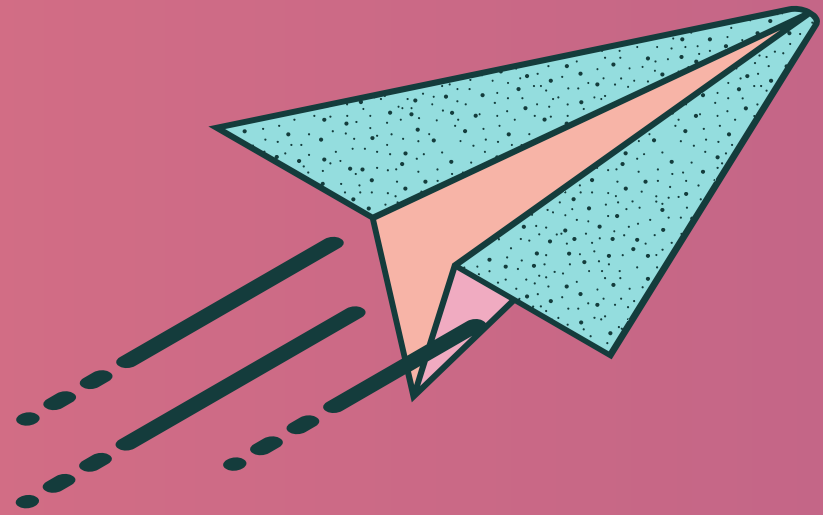
4. Pixel Copying Strategy + Inpainting

- Copies pixels from adjacent regions to fill blanks
- Prioritizes left-right neighbors, uses top-bottom as backup

Conclusions

- This study presents an automated image processing framework combining YOLO-based corruption detection with variety of image restoring algorithms.
- By utilizing YOLO, we enabled the precise detection and localization of various image corruption types.
- Employ DIP structure, utilizing inplainting to deal with block corruption and generate the missing part according to near pixels.
- Employ DIP structure, applying super resolution model to improve blurred images to deal with ghosting and mosaic corruptions.
- resolve tear corruption by utilizing Image registration with inplainting technology

Overall, this research confirms the potential of using object detection for image quality diagnosis, offering a robust basis for the development of more intelligent and adaptive image restoration systems.



Reference

Classify

https://blog.csdn.net/weixin_44445800/article/details/144595026

Deep Image Prior

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<https://reurl.cc/W8NQ99>



Thank You For Your Attention

